



U.S. AIRLINE PERFORMANCE & DELAY ANALYSIS — 2015

A Comprehensive Data Analytics Report

Prepared By

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MySQL 8.0 | Python | Power BI Desktop

Dataset: 5,819,079 Flights | 14 Airlines | Year: 2015

1. Introduction & Problem Statement

The aviation industry is one of the most operationally complex sectors, where even minor inefficiencies cascade into significant economic and passenger-experience costs. In 2015, the U.S. domestic aviation network processed nearly 5.82 million flights operated by 14 major carriers — yet a substantial portion of those flights experienced delays, cancellations, or diversions.

This report presents a comprehensive analysis of U.S. domestic flight performance data for the full calendar year 2015. The analysis was conducted using a structured data pipeline: raw data was ingested and transformed in MySQL, analytical queries were executed to extract insights, and an interactive Power BI dashboard was developed for visual exploration.

Problem Statement:

Despite advances in scheduling technology, U.S. airlines continue to suffer from high delay and cancellation rates. Stakeholders — including airline operations teams, airport authorities, and passengers — lack consolidated, actionable visibility into the root causes, timing patterns, and airline-level performance differentials driving these disruptions. This analysis addresses that gap.

2. Objectives

This project was designed to answer the following analytical questions:

- Which airlines deliver the best and worst on-time performance (OTP)?
- What are the primary causes of flight delays and cancellations?
- Which months, days of week, and time-of-day slots carry the highest delay risk?
- Which routes and airports are most prone to delays and cancellations?
- What proportion of delays are within airline control vs. external factors?
- How do delay patterns vary across the calendar year?

The ultimate goal is to surface actionable recommendations for airlines, airports, and passengers to reduce disruptions and improve punctuality.

3. Methodology

3.1 Data Sources

The analysis is based on the 2015 U.S. Flight Delays and Cancellations dataset from the Bureau of Transportation Statistics (BTS), publicly available via Kaggle. It comprises three CSV files:

File	Contents	Rows / Records
flights.csv	Flight-level operational data (delays, times, distances)	5,819,079 flights
airlines.csv	IATA codes and full airline names	14 airlines
airports.csv	Airport codes, cities, states, coordinates	322 airports

3.2 Tools & Technology Stack

Tool	Purpose
MySQL 8.0	Database creation, data ingestion, staging, transformation, indexing, views
Python	Bulk loading of large flights.csv file (5.8M rows) via chunked inserts
Power BI Desktop	Interactive dashboard — 6 pages covering overview, delay, route, time, cancellation, and conclusion

3.3 Analytical Approach

The workflow followed four phases:

- Data Ingestion & Staging: Raw CSVs loaded into a staging table with all-TEXT columns to avoid import errors, then cast and inserted into the production flights table.
- Data Transformation: Derived columns added — FLIGHT_DATE (DATE), SCHEDULED_DEP_TIME / ACTUAL_DEP_TIME / SCHEDULED_ARR_TIME / ACTUAL_ARR_TIME (TIME), DEPARTURE_HOUR (TINYINT), IS_DELAYED (binary flag: arrival delay > 15 min and not cancelled), and CANCELLATION_REASON_DESC (human-readable label). NULL delay values for non-cancelled flights were set to 0.
- Indexing & Views: Nine indexes created on high-cardinality filter/group-by columns. A comprehensive view (vw_flights_enriched) joins flights with airlines and airports for enriched querying.
- Analysis & Visualisation: SQL queries computed OTP rates, delay breakdowns by cause, monthly/daily/hourly trends, airline rankings, and route-level metrics. Results were connected to Power BI for dashboard construction.

A flight is classified as delayed if its arrival delay exceeds 15 minutes (the FAA standard) and it was not cancelled.

4. Key Findings at a Glance

5.82M Total Flights 2015	1.02M Delayed Flights 17.6% of flights	89,884 Cancelled Flights 1.54% rate	82.14% Overall OTP Rate On-Time Performance	4.41 min Average Arrival Delay All flights incl. early
89.47% Best OTP Airline Hawaiian Airlines	71.25% Worst OTP Airline Spirit Air Lines	Oct Best OTP Month 88.2% OTP	Mar Worst Delay Month 9.6 min avg delay	72% Controllable Delays Airline + Late Aircraft

These headline numbers reveal a system where roughly 1 in 6 operated flights arrived more than 15 minutes late, and the vast majority of delay minutes are attributable to factors within airline control — highlighting significant scope for operational improvement.

5. Airline Performance Analysis

The table below presents complete performance data across all 14 carriers for 2015. Airlines are sorted by OTP rate (descending).

Airline	Total Flights	Delayed	OTP Rate %	Cancel Rate %	Average Arr Delay (min)
Hawaiian Airlines	76,272	8,010	89.47	0.22	2.02
Alaska Airlines	172,521	21,162	87.69	0.39	-0.98
Delta Air Lines	875,881	113,112	87.03	0.44	0.19
American Airlines	725,984	125,238	82.49	1.50	3.45
US Airways	198,715	34,930	82.05	2.05	3.71
SkyWest Airlines	588,353	103,850	82.05	1.69	5.85
Southwest Airlines	1,261,855	227,317	81.75	1.27	4.37
Virgin America	61,903	11,348	81.51	0.86	4.74
Atlantic Southeast	571,977	105,331	81.08	2.66	6.59
United Air Lines	515,723	101,303	80.10	1.27	5.43
American Eagle	294,632	58,623	79.03	5.10	6.46
JetBlue Airways	267,048	57,257	78.21	1.60	6.68
Frontier Airlines	90,836	22,851	74.68	0.65	12.50
Spirit Air Lines	117,379	33,166	71.25	1.71	14.47

System Total | 5,819,079 flights | 82.14% OTP | 1.54% cancellation rate | 4.41 min average delay

5.1 Key Insights

- Hawaiian Airlines leads on OTP (89.47%) with the lowest cancellation rate (0.22%) — benefiting from short, controlled inter-island routes and stable Pacific weather.
- Alaska Airlines and Delta Air Lines are the standout large-network performers, each achieving OTP above 87.69%, well above the system average of 82.14%.
- Southwest Airlines operated the highest flight volume (1.26M flights) while maintaining a solid 81.75% OTP — demonstrating strong operational efficiency at scale.
- Spirit Air Lines recorded the worst OTP (71.25%) and the highest average arrival delay (14.47 min), reflecting structural operational challenges in its ultra-low-cost model.
- American Eagle Airlines had the highest cancellation rate (5.10%) — nearly 3.3x the system average — driven partly by its regional turboprop operations susceptible to weather and NAS congestion.

6. Delay Analysis

6.1 Delay Causes Breakdown

Delay minutes were attributed across five official FAA categories. The Delay Causes donut chart in the dashboard shows:

Delay Category	Share of Total Delay Minutes	Nature
Late Aircraft (Cascading)	39.84%	Controllable — prior flight arrived late
Airline / Carrier	32.20%	Controllable — maintenance, crew, turnaround
Air System (NAS)	22.88%	Partially controllable — ATC, runway congestion
Weather	4.95%	External / uncontrollable
Security	0.13%	External / uncontrollable

Key insight: 72% of all delay minutes (Late Aircraft + Airline) are directly controllable by airlines. Weather the most visible and emotionally attributed cause accounts for less than 5% of delay minutes.

6.2 Monthly Delay Trend

Average arrival delay varied substantially by month. The table below matches the Monthly Delay Trend line chart in the Delay Analysis dashboard:

Month	Average Arrival Delay (min)	Note
January	5.81	Winter weather impacts begin
February	6.10	Elevated — winter storms; highest cancellation volume (4.78%)
March	9.60	Peak delay month — late-winter / spring weather transition
April	-0.8	Best month for delays (most flights arrive early)
May	3.20	Stable — spring shoulder season
June	8.30	Second-highest delay month — summer travel surge begins
July	4.60	Moderate — mid-summer
August	4.90	Moderate — late summer
September	4.50	Stable — post-summer demand ease
October	1.10	Second-best delay month; best OTP month (88.2%)
November	6.40	Rises again — early holiday travel pressure
December	-0.8	Early flights mask end-of-year pressure

March (9.6 min) and June (8.3 min) represent the two peaks of delay severity. April and December both record the lowest average delay (-0.8 min). October delivers the best combined performance low delays (1.1 min) alongside the highest OTP rate (88.2%).

7. Route & Airport Analysis

7.1 Top Busiest Routes

Route	Origin City	Destination	Flights	Average Arr Delay (min)
SFO → LAX	San Francisco	Los Angeles	13,744	11.44
LAX → SFO	Los Angeles	San Francisco	13,457	10.74
JFK → LAX	New York	Los Angeles	12,016	-2.67
LAX → JFK	Los Angeles	New York	12,015	0.61
LAS → LAX	Las Vegas	Los Angeles	9,715	11.97
LGA → ORD	New York	Chicago	9,639	3.77
LAX → LAS	Los Angeles	Las Vegas	9,594	8.66

7.2 Highest-Delay Routes

Origin	Destination	Average Arr Delay (min)	Delay Rate %
Aspen	Dallas-Fort Worth	39.97	33%
New York	Eagle (CO)	38.01	30%
Atlantic City	Detroit	30.90	41%
Minneapolis	Trenton	30.23	42%
Atlantic City	Boston	29.42	42%
Eagle (CO)	Chicago	28.93	27%
Dallas-Fort Worth	Kahului	28.86	38%

7.3 Top Airports by Flight Volume

Origin City / Airport	Total Departures
Chicago (O'Hare + Midway)	366,770
Atlanta (Hartsfield-Jackson)	346,836
Dallas-Fort Worth	239,551
Houston	198,664
Denver	196,055
Los Angeles	194,673
New York (JFK + LGA)	193,416
San Francisco	148,008
Phoenix	146,815

Chicago leads all cities with 366,770 departures — and carries some of the highest delays due to O'Hare's congestion and Midwest weather exposure.

8. Cancellation Analysis

Of the 5,819,079 total flights, 89,884 were cancelled — a system-wide rate of 1.54%.

8.1 Cancellation by Cause

Cancellation Reason	Count	Share of Cancellations
Weather	48,851	54.35% — dominant cause
Airline / Carrier	25,262	28.11%
National Air System	15,749	17.52%
Security	~18	0.02%

Weather dominates cancellations (54.35%), reflecting the vulnerability of the U.S. network to winter storms (Jan–Feb) and convective weather (June). In contrast, weather accounts for only ~5% of delay minutes — suggesting airlines absorb weather impact into delays wherever possible, resorting to cancellations only when flights are entirely non-viable.

8.2 Monthly Cancellation Pattern

Month	Cancellation Rate %	Note
January	2.55%	High — winter storms
February	4.78%	PEAK — worst cancellation month; late-winter storms
March	2.18%	Easing — spring weather improving
April	0.93%	Lowest month — stable spring weather
May	1.15%	Low — stable
June	1.81%	Rises — summer convective weather
July	0.92%	Low — summer peaks past
August	0.99%	Low — stable
September	0.45%	Low — post-summer
October	0.50%	Second-lowest — best combined OTP + cancel window
November	0.98%	Moderate rise — early winter travel begins
December	1.68%	Higher — winter weather returns; year-end holiday pressure

Cancellation rate peaked in February at 4.78% — the single worst month — driven by late-winter storms. January (2.55%) is the second worst. The lowest rates occur in September (0.45%) and October (0.5%).

8.3 Airlines & Cities with Highest Cancellation Rates

Airlines:

- American Eagle Airlines Inc. — 5.10% (highest; regional operations most exposed)
- Atlantic Southeast Airlines — 2.66%
- US Airways — 2.05%

Most-cancelled origin cities:

- Ithaca, NY — 11.76% cancellation rate
- Mammoth Lakes, CA — 10.26%
- Hailey (Sun Valley), ID — 9.21%
- Devils Lake, ND — 8.76%
- Aspen, CO — 7.75%

Small regional airports serviced by turboprop commuter operations face disproportionately high cancellation rates due to weather sensitivity and thinner scheduling buffers.

9. Time Trend Analysis

9.1 Day of Week

Day	Delay Rate %	Assessment
Monday	19%	Worst — business travel peak + weekend recovery
Thursday	19%	Joint worst — end-of-week travel surge
Friday	18%	High — weekend leisure travel
Sunday	17%	Elevated — return travel
Tuesday	17%	Moderate
Wednesday	17%	Moderate
Saturday	15%	Best — lowest delay rate

Saturday stands out as the most reliable travel day with a 15% delay rate — 4 percentage points better than the worst days. Business travellers and leisure flyers would benefit from Saturday departures where feasible.

9.2 Departure Hour Analysis

Delays accumulate throughout the day as aircraft cycles compound (the 'ripple effect'):

- 05:00–08:00: Lowest average arrival delay (negative — flights arrive early). Best time slot for punctuality.
- Midday (10:00–14:00): Moderate delays building as cascading effects accumulate.
- Late afternoon to evening (16:00–21:00): Peak delay window, exceeding 10 min average delay by 20:00–21:00.

Early morning departures benefit from aircraft that have not yet accumulated cascading delays. Passengers prioritising punctuality should book the first departure of the day.

10. Power BI Dashboard Walkthrough

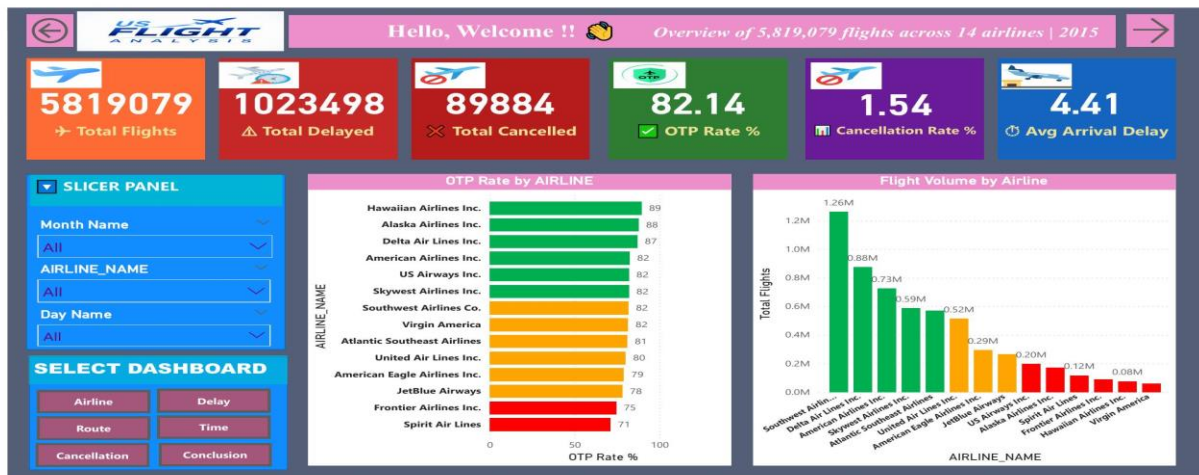
The Power BI dashboard consists of six interactive pages plus a cover page and conclusion, each targeting a specific analytical dimension. All pages include Month Name, Airline Name, and Day Name slicers for cross-filtering.

Page	Title	Key Visuals
Cover	Dashboard Entry	Full-screen visual; 'Enter Dashboard' bookmark button navigates to Overview
1	Overview	6 KPI cards; OTP Rate by Airline bar; Flight Volume by Airline bar; Slicer Panel; Select Dashboard navigation
2	Airline Performance	Airline logo slicers; Average Arrival Delay bar; OTP vs Delay Rate scatter; Full performance summary table
3	Delay Analysis	Delay Causes donut; Monthly Delay Trend line; Delay Rate by Airline bar; Monthly OTP Heatmap
4	Route & Airport Analysis	Total Flights map; Top Airports by volume; Top Airports by average delay; Busiest Routes; Highest-Delay Routes
5	Time Trends	Monthly OTP line; Delay Rate by Day of Week; Average Delay by Departure Hour; Monthly Delay vs Cancellation
6	Cancellation Analysis	Cancellation Reasons donut; Monthly Trend; Cancel Rate by Airline; Most Cancelled Cities; by Month stacked bar
7	Conclusion	Best/Worst OTP airline, month, day, time slot; top airport; most cancelled airline; 5 headline findings

Dashboard Page — Cover Page



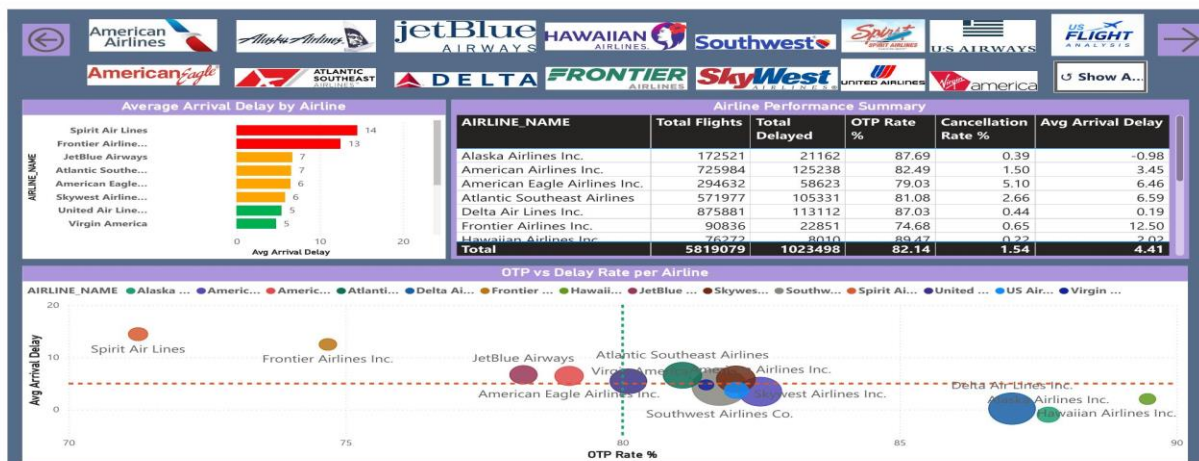
Dashboard Page 1 — Overview (KPIs, OTP Rate, Flight Volume)



Key Insights

- Overall, On-Time Performance (OTP) stood at 82.14%, indicating that over four out of five flights arrived on time.
- Total delayed flights exceeded 1.02 million, while cancellations accounted for 89,884 flights.
- Hawaiian Airlines achieved the highest OTP rate (89%), whereas Spirit Airlines recorded the lowest OTP (71%).
- Southwest Airlines operated the highest number of flights, making it the largest carrier.

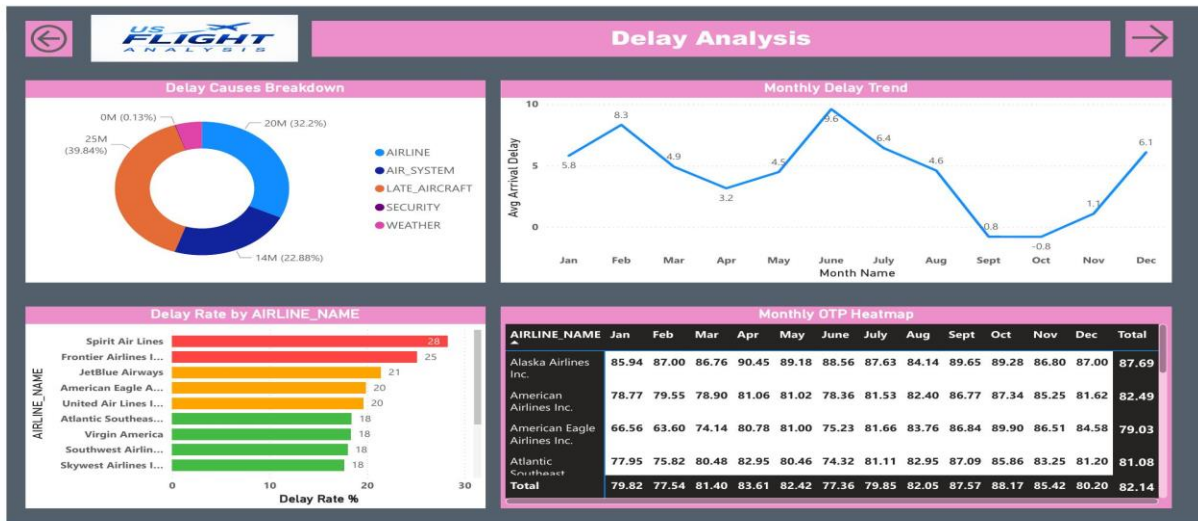
Dashboard Page 2 — Airline Performance (Logos, Delay Bar, OTP Scatter, Summary Table)



Key Insights

- Spirit Airlines** and **Frontier Airlines** experienced the highest average arrival delays, exceeding 12 minutes.
- Delta Air Lines** maintained high punctuality while operating a large flight network, indicating strong operational efficiency.
- American Eagle Airlines** recorded the highest cancellation rate (5.10%), significantly above the overall average.
- Airlines with higher OTP generally showed lower average arrival delays, confirming a strong inverse relationship between punctuality and delays.

Dashboard Page 3 — Delay Analysis (Causes Donut, Monthly Trend, OTP Heatmap)



Key Insights

- **Late Aircraft Delay** was the largest contributor, accounting for **39.84%** of total delay minutes.
- Together, airline-related and late-aircraft delays represented over **72% of total delays**, suggesting significant opportunities for operational improvement.
- Delay severity peaked during **March and June**, while **April and December** showed the lowest average delays.
- **Spirit Airlines** recorded the highest delay rate, followed by **Frontier Airlines** and **JetBlue Airways**.

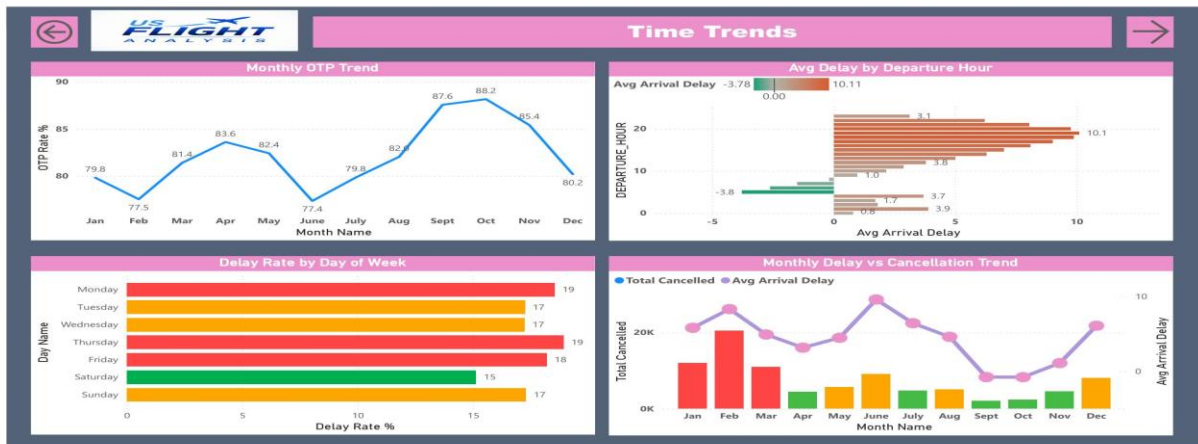
Dashboard Page 4 — Route & Airport Analysis (Map, Top Routes, Delay Routes)



Key Insights

- The busiest routes were concentrated between major metropolitan hubs, particularly **San Francisco (SFO) ↔ Los Angeles (LAX)** and **New York (JFK) ↔ Los Angeles (LAX)**.
- **Chicago** emerged as the busiest airport by flight volume, followed by **Atlanta** and **Dallas-Fort Worth**.
- Certain smaller airports such as **Gustavus** and **Pago Pago** experienced the highest average arrival delays, indicating localized operational challenges.
- Routes involving **Aspen**, **Atlantic City**, and **Trenton** exhibited significantly higher delay rates than the network average.

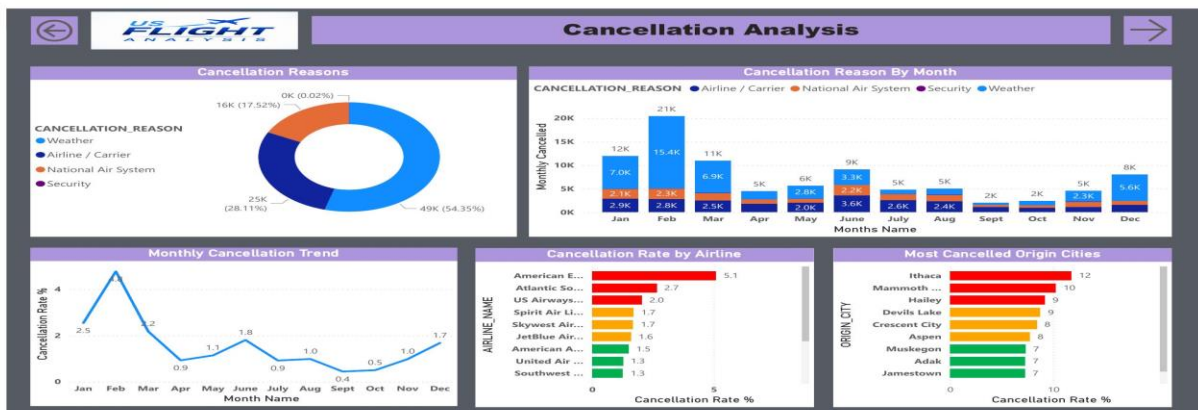
Dashboard Page 5 — Time Trends (Monthly OTP, Day of Week, Departure Hour)



Key Insights

- October** recorded the highest OTP performance, while overall punctuality improved during the latter part of the year.
- Monday** and **Thursday** experienced the highest delay rates (19%), making them the most delay-prone weekdays.
- Saturday** recorded the lowest delay rate (15%), making it the most reliable day for travel.
- Flights departing during **early morning hours (05:00–08:00)** experienced the lowest arrival delays.

Dashboard Page 6 — Cancellation Analysis (Reasons, Monthly Trend, Cities)



Key Insights

- Airline/Carrier-related issues** accounted for the largest share of cancellations (54.35%), highlighting operational inefficiencies as the primary cancellation driver.
- Weather-related disruptions** contributed 28.11% of cancellations, making weather the second most significant factor.
- February** recorded the highest cancellation rate (4.78%), while **September** showed the lowest.
- American Eagle Airlines** experienced the highest airline-level cancellation rate (5.10%).
- Airports such as **Ithaca**, **Mammoth Lakes**, and **Hailey** reported the highest cancellation rates among origin cities.

Dashboard Page 7 — Conclusion (Best / Worst Summary, Key Findings)

CONCLUSION			
Best OTP Airline Hawaiian Airlines (89.4%)	1) October recorded the highest OTP, with Saturday and the 05:00–08:00 time slot delivering the best punctuality. 2) Hawaiian Airlines emerged as the top-performing carrier with the highest reliability and lowest cancellation rate, while Southwest Airlines operated the most flights. 3) Chicago O'Hare was the busiest airport with highest delays. 4) Weather was the leading cause of cancellations (54.35%). 5) 72% of all delay minutes are controllable by airlines.		
Worst OTP Airline Spirit Air Lines (71.1%)			
Best Month October (88.2% OTP)	Top Airport Chicago Total Flight (366770)	Most Cancelled Airline American Eagle Airlines Inc. Cancellation Rate (5.10%)	
Worst Month February (4.78% cancel)	Best Day Saturday Delay Rate (15%)	Worst Day Thursday Delay Rate (19%)	Best Time Slot 05:00 - 08:00 (Lowest Arrival Delay)

Key Insights

- **Hawaiian Airlines** emerged as the best-performing airline with the highest OTP and strong operational reliability.
- **Spirit Airlines** showed the weakest operational performance due to high delays and low OTP.
- More than **72% of delay minutes were controllable by airlines**, suggesting that operational improvements could substantially enhance punctuality.
- **Chicago** was the busiest airport in the network, serving as a critical aviation hub.
- Travelers seeking the most reliable experience should prefer **Saturday morning departures**, particularly during the **05:00–08:00** time window.

11. Actionable Recommendations

For Airlines

- Reduce late-aircraft cascades: 39.84% of delay minutes stem from prior-flight lateness. Build buffer time into mid-day and evening rotation schedules at Chicago, Atlanta, and Dallas-Fort Worth.
- Invest in turnaround efficiency: The 32.2% airline-controlled delay share points to maintenance and crew readiness gaps. Predictive maintenance analytics and crew scheduling optimisation can directly reduce this.
- Focus on bottom-quartile carriers: Spirit Air Lines, Frontier, and JetBlue — OTP rates of 71–78% — have the greatest improvement opportunity.
- American Eagle cancellation reduction: With a 5.10% cancellation rate, route-level weather-risk assessment and proactive re-accommodation protocols are essential.

For Airport Authorities

- Chicago O'Hare capacity management: As the busiest and most delay-prone hub, investment in runway throughput and gate management would generate system-wide benefits.
- Regional airport resilience: Airports like Ithaca, Mammoth Lakes, and Aspen (above 8% cancellation) need improved ground handling and de-icing infrastructure.

For Passengers

- Book early morning departures (05:00–08:00) — average arrival delays in this window are negative (flights arrive early on average).
- Choose Saturdays and avoid Mondays/Thursdays for the lowest delay risk.
- Prefer October, April, or September for travel. Avoid March (worst delay month) and February (worst cancellation month).
- Select Hawaiian, Alaska, or Delta for the highest reliability.

12. Conclusion & Limitations

Conclusion

This analysis of 5.82 million U.S. domestic flights in 2015 reveals a system performing at an 82.14% on-time rate — with substantial variance across airlines, routes, seasons, and departure times. The most significant finding is that 72% of delay minutes are airline-controllable, meaning the industry has considerable power to improve performance without waiting for weather conditions to change.

Hawaiian Airlines demonstrated that high OTP is achievable at scale, while Southwest Airlines proved that the highest-volume operator can maintain competitive punctuality. October is the standout month for reliability, early-morning departures are strongly preferable, and Saturday is the safest day to fly. Weather dominates cancellation causes (54.35%) but contributes only ~5% of delay minutes.

The Power BI dashboard provides stakeholders with an interactive, drill-down tool to explore these findings at any level of granularity — from system-wide KPIs to specific airline-month-route combinations.

Limitations

- Data vintage: The dataset covers 2015 only. Post-COVID structural changes may mean findings do not fully apply to the current environment.
- Causal inference: The analysis identifies correlations but cannot establish causal chains.
- Missing data: A small subset of records had data quality issues handled via NULL exclusions.
- Passenger impact not modelled: Connection disruptions, compensation costs, and load factors are not captured.
- External cost data absent: Economic cost of delays could not be quantified without additional revenue/cost datasets.

THANK YOU